

Student Dropout Prediction: A Machine Learning Analysis

DSAA2011(L01) Course Project - HKUST(GZ) 2026 Spring

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Part A

Clustering (1/2): Algorithm Selection & Tuning

1. Hard-Boundary Models

- ▶ **Exploration:** Tested Hierarchical (Ward) and K-Means.
- ▶ **Optimal K:** $K = 4$ provided the best granularity.
- ▶ **Limitation:** While K-Means performed well, rigid spherical boundaries are less flexible in handling the severe structural overlap of the "Enrolled" class.

⇒ *Conclusion: A probabilistic approach is needed to handle the overlap.*

2. Final Selection: GMM

- ▶ **Best Config:** $K = 4$ with **Tied Covariance**.
- ▶ **Why GMM?** Soft-assignments handled overlap naturally; "Tied" covariance reduced 40D overfitting.

Performance Comparison ($K = 4$)

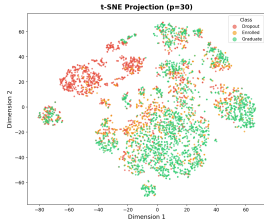
Algorithm	ARI	NMI
Hierarchical (Ward)	0.153	0.149
K-Means	0.313	0.230
GMM (Tied)	0.334	0.273

Clustering (2/2): Core Insight & Visualizations

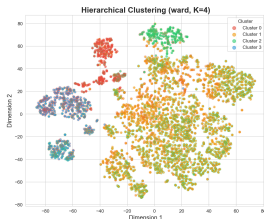
Insight: Natural Label Consistency

- **Consistency:** Clustering results seem to be basically consistent with true labels.
- **Ablation Check:** Removing target-encoded features yielded identical structures.

Original Labels



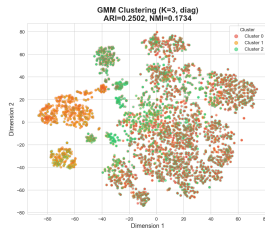
Hierarchical (Ward)



K-Means ($K = 4$)



GMM (Tied)



Part C

Part D

Reference

Realhino, V., et al. (2022). Student Dropout and Academic Success. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>

Q & A

Thank You!